

Track A: Depth Estimation & Camera Geometry

Team: 1 human + 1 AI agent **Owns:** `/track_a_depth` **You are first in the pipeline.** You have no upstream dependency, so you start on real data Day 1 — but your first job is producing the mock fixtures everyone else needs (see §2 below) before you go deep on your own track.

1. What this track delivers

For any input photo, produce: - `depth.npy` — a per-pixel depth map from Depth Anything V2 (pretrained, untouched — no training in this MVP) - `intrinsic.json` — an estimated camera focal length / field of view for that photo

Both in the exact shapes defined in `00_shared_contract.md` §2. That contract is not negotiable without telling the other three tracks.

2. Day 1, first task: build the mock fixtures

Before touching the real model, produce `shared/sample_jobs/job_0001/`: a fake `depth.npy` (e.g. a smooth gradient with 2-3 synthetic “blobs” of different depth) and a fake `intrinsic.json` with plausible numbers. This is what unblocks Track B and Track C to start the same day instead of waiting on your real model to work. Takes under an hour; do it first.

3. Who does what, and why

This is the track where the split between human and agent matters most, because depth models are compute- and environment-heavy in ways an agent alone cannot execute.

What the agent does: - Writes the loading/inference code for Depth Anything V2 (model selection, checkpoint download, pre/post-processing, output formatting to match the contract). - Writes the FOV-based intrinsic estimation code and explains the math behind it (pinhole camera model, how FOV assumption maps to fx/fy). - Debugs errors, reviews the human’s code, and explains *why* a given depth map looks wrong (e.g. flipped near/far, wrong normalization, checkpoint mismatch) rather than just handing over a fix. - Proposes what to try next when accuracy looks off, and why — e.g. “this looks like the outdoor checkpoint on an indoor scene, try the indoor variant” — turning the human’s raw observation into a concrete next step.

What the human does (things the agent structurally cannot do): - Has the actual GPU/local machine — downloads the multi-GB checkpoints, runs inference locally, reports back real runtimes and errors from their own environment. - Takes or sources the real test photos (the 5-10 “hero images”), including photos of objects with **known real-world dimensions** — this is required later for Track D’s accuracy validation, and only a human can go measure a real object with a tape measure and photograph it. - Visually judges output quality: does this depth map actually look like the scene? A flat wall reading as “near” in one corner and “far” in another is something a human eye catches instantly and a script won’t unless told exactly what to check for. - Makes the indoor-vs-outdoor variant call per test image, and reports it back so the agent can wire the branch logic correctly.

How they work together, concretely: the human runs the agent’s code on real photos, screenshots/describes what comes out, and the agent reads that description to diagnose the next fix. Neither role is optional — the agent without the human has no way to execute or see real output; the human without the agent has no fast way to write and debug the inference/geometry code.

4. Day-by-day

Day	Task
1	Mock fixtures (above) → published to <code>/shared/sample_jobs</code> . Then: agent scaffolds model loading + inference code; human downloads checkpoint and runs first real inference locally.
2	Get depth output matching <code>depth.npy</code> contract shape/dtype on 1 real photo. Human visually sanity-checks it; agent adjusts normalization/format based on human's description.
3	Agent implements FOV-based intrinsics estimation; human tests it against 2-3 photos of objects with known focal length (phone EXIF data, if available, is a useful cross-check the human can pull).
4	Run all 5-10 hero images through the full Track A pipeline. Human flags any that look wrong; agent triages and, where time allows, fixes. Write the "what we actually built vs. contract" note.
5	Integration day — plug real <code>depth.npy</code> / <code>intrinsics.json</code> into Track B's real pipeline (not mocks) alongside the other pairs.

5. Definition of done

- Real `depth.npy` + `intrinsics.json` produced for all hero images, in the exact contract format.
- Mock fixtures published Day 1 (unblocked Tracks B/C/D).
- One paragraph documenting any deviation from the contract format, if any.

6. If you get stuck

Ask the agent to explain the *why*, not just paste a fix — the human on this track will be the one debugging checkpoint/environment issues alone during Day 5 integration if the agent isn't there, so understanding beats speed here.